

Entamoeba histolytica (intestinal & tissue ameba)

Microbiology department

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Etiology

- Amebiasis is a parasitic infection caused by the protozoal organism *E histolytica* found worldwide.
- Global Impact: It affects millions annually, especially in tropical regions with poor sanitation. Estimated 40,000–100,000 deaths occur worldwide each year.

Causative Agent: *Entamoeba histolytica*.

Forms:

- **Cyst:** Infectious, dormant form.
- **Trophozoite:** Active, disease-causing form.
- **Transmission:** Fecal-oral route via contaminated food, water, or surfaces

- The genus *Entamoeba* contains many species, (*E histolytica*, *Entamoeba dispar*, *Entamoeba moshkovskii*, *Entamoeba bangladeshi*, *Entamoeba polecki*, *Entamoeba coli*, and *Entamoeba hartmanni*) can reside in the human interstitial lumen.
- *E histolytica* is the only one definitely associated with disease, *E moshkovskii* to be the causative agent of diarrhea in infants.
- Although *E dispar* and *E histolytica* cannot be differentiated by means of direct examination, molecular techniques have demonstrated that they are indeed two different species, with *E dispar* being commensal (as in patients with HIV infection) and *E histolytica* pathogenic.

Entamoeba histolytica

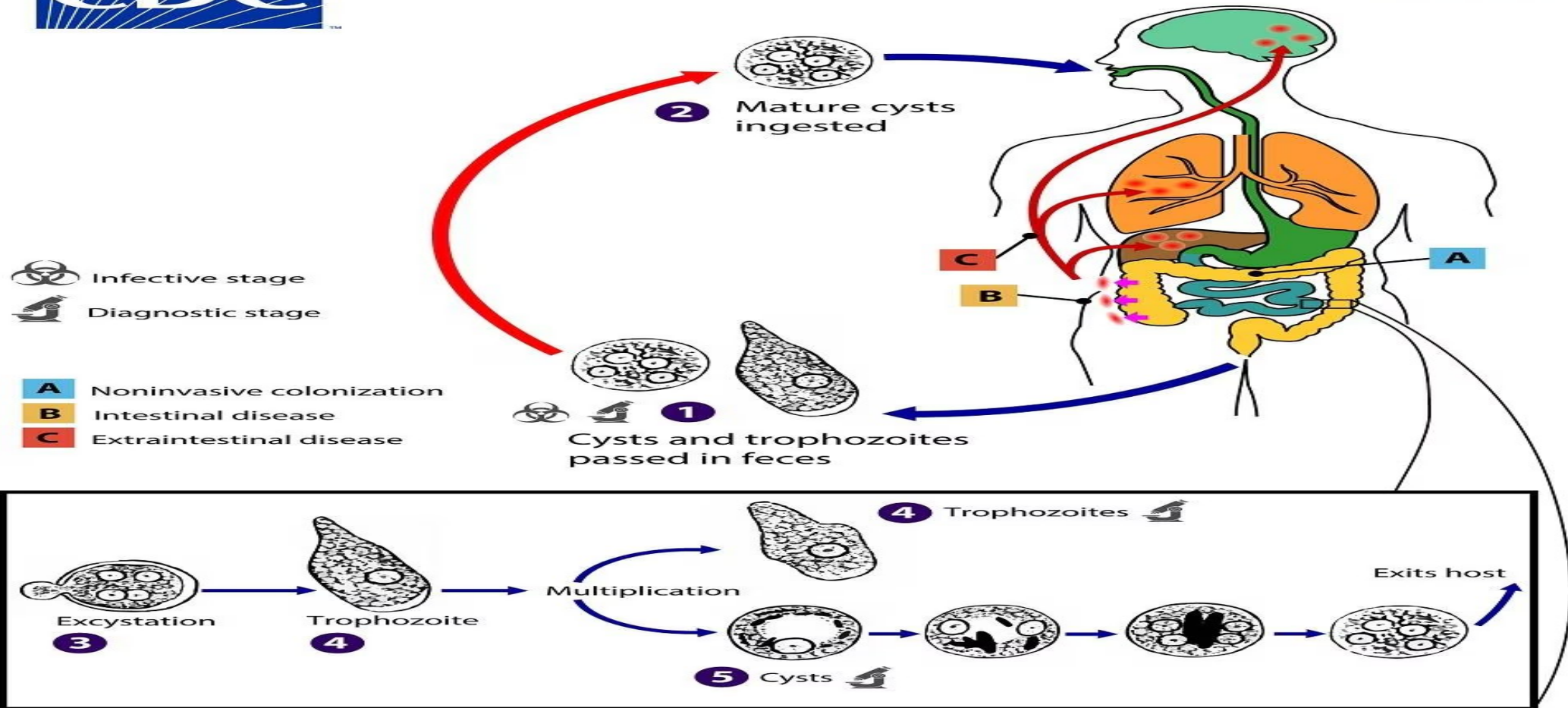
- Cysts are present in the lumen of the colon and in mushy or formed feces. They contain glycogen vacuole & chromatoid bodies.
- Nuclear division occurs within the cyst, resulting quadri-nucleated cyst.
- Diagnosis in most cases results on the characteristics of the cyst, as trophozoites usually appear in diarrheic feces in active cases & survive only for few hours & in tissues.
- The cytoplasm has two zones, a hyaline outer margin & a granular inner region that may contain red blood cells (pathogenomic), the nuclear membrane is lined by fine, regular granules of chromatin with a small central body (endosome or charyosome).

Transmission

- Fecal-oral route.
- Contaminated food and water supplies and contaminated hands of food handlers by infective cysts in feces.
- Sexual transmission, especially in the setting of oral-anal practices (anilingus).
- Poor nutrition, through its effect on immunity, has been found to be a risk factor for amebiasis.



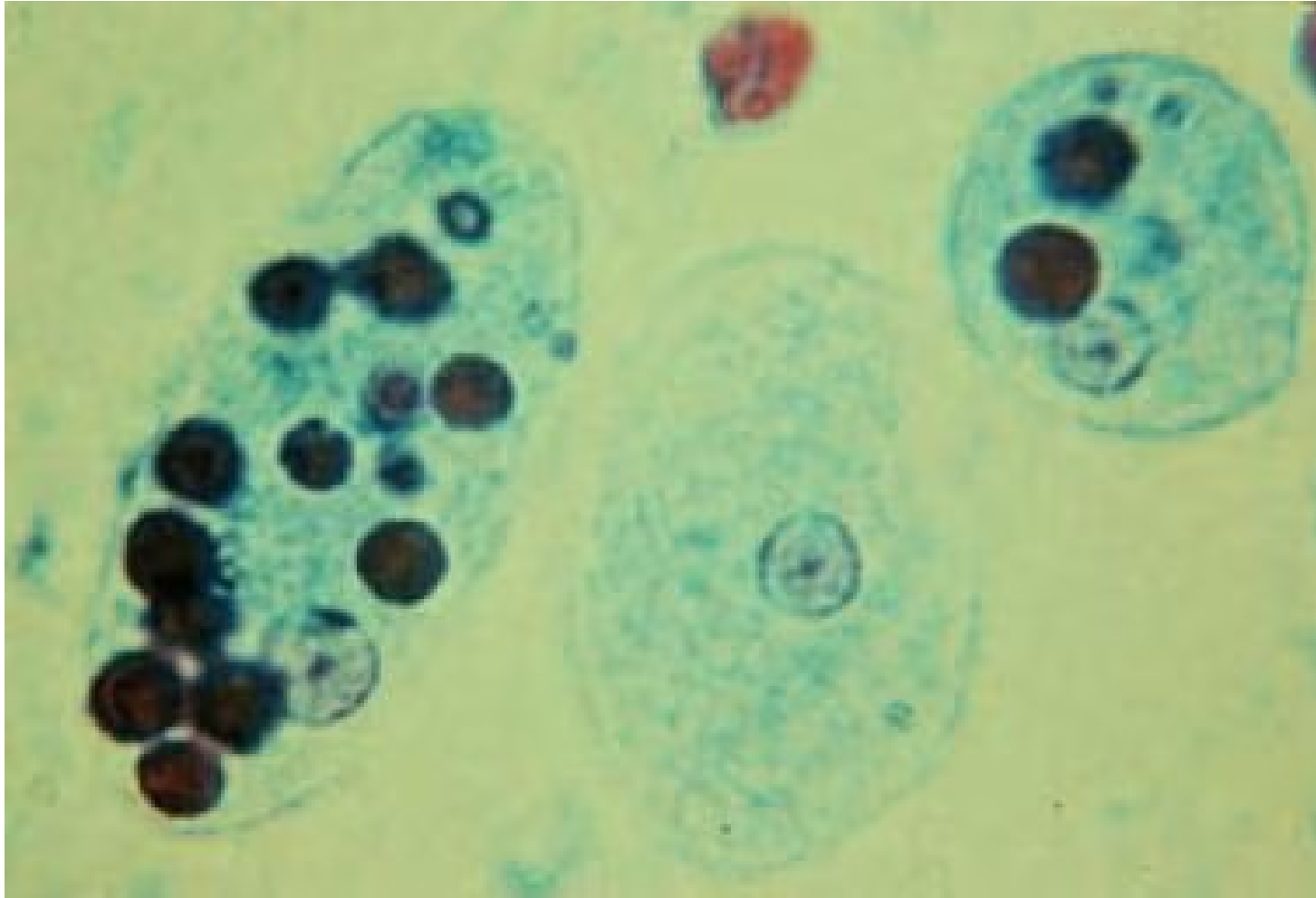
Amebiasis



Life cycle of amebiasis (*Entamoeba histolytica*). Courtesy of the Centers for Disease Control and Prevention (CDC).

Pathophysiology

- *E histolytica* is a pseudopod-forming, non-flagellated protozoal parasite that causes proteolysis and tissue lysis and can induce host-cell apoptosis. Humans and perhaps non-human primates are the only natural hosts.
- Ingestion of *E histolytica* cysts from the environment is followed by excystation in the terminal ileum or colon to form highly motile trophozoites.
- Upon colonization of the colonic mucosa, the trophozoite may encyst and is then excreted in the feces, or it may invade the intestinal mucosal barrier and gain access to the bloodstream, whereby it is disseminated to the liver, lung, and other sites. Excreted cysts reach the environment to complete the cycle.
- Disease caused by only a small number of cysts.



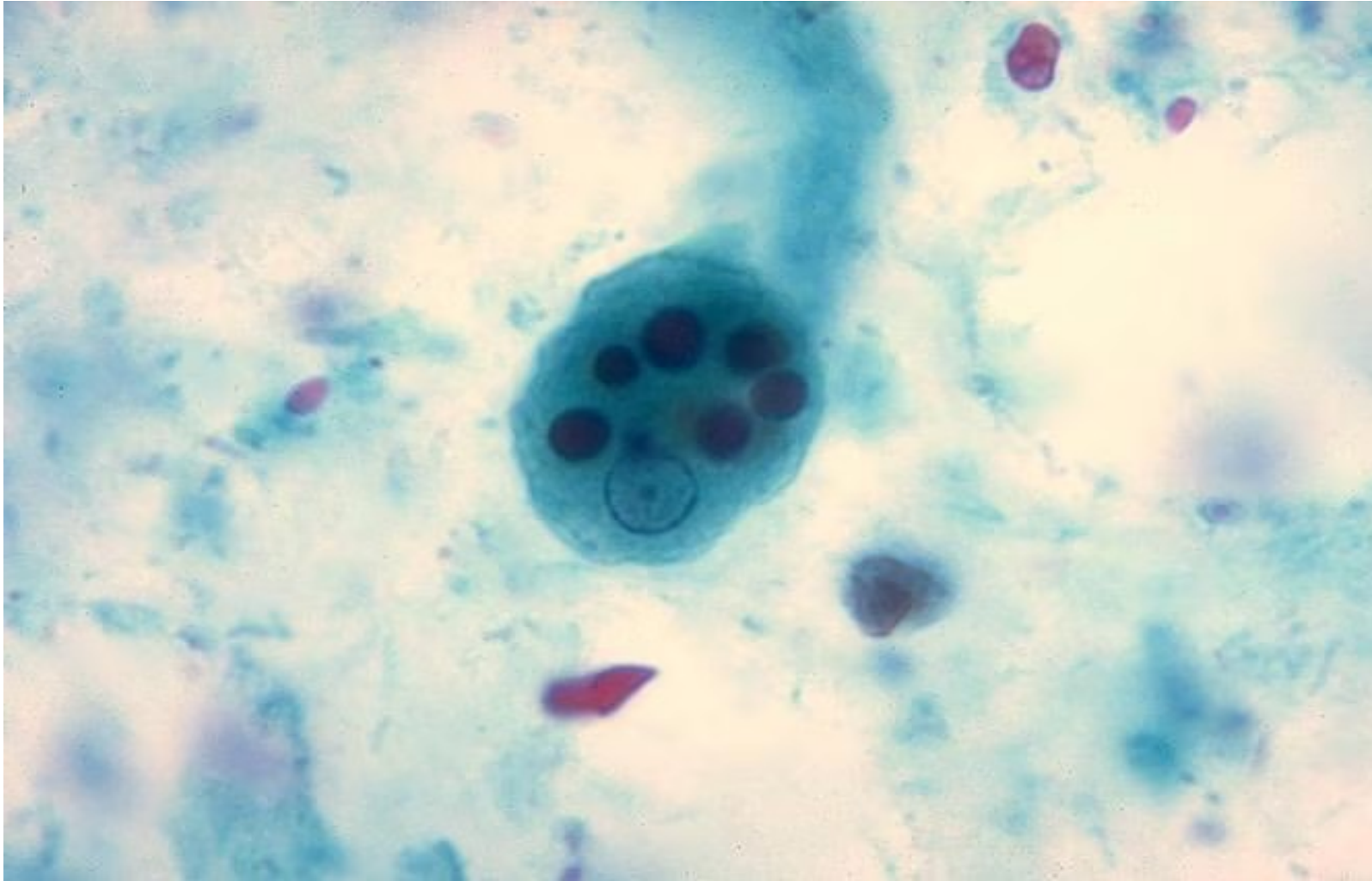
Trichrome stain of ***Entamoeba histolytica*** trophozoites in amebiasis. Two diagnostic characteristics are observed. Two trophozoites have ingested erythrocytes, and all 3 have nuclei with small, centrally located karyosomes.

Clinical Manifestations

- **Intestinal Amebiasis:**
 - Diarrhea (may be bloody)
 - Abdominal pain
 - Cramping
 - Fever
- **Extraintestinal Amebiasis:**
 - Liver abscess (most common)
 - Pleuropulmonary involvement
 - Rarely, brain abscess



Entamoeba histolytica cyst. Image courtesy of Centers for Disease Control and Prevention.



Entamoeba histolytica trophozoite. Image courtesy of Centers for Disease Control and Prevention.

Adherence and Immune Response

- The adherence of trophozoites to colonic epithelial cells seems to be mediated by a galactose/N-acetylgalactosamine (GAL/GalNAc)–specific lectin, a 260-kd surface protein containing a 170-kd subunit and a 35-kd subunit.
- A mucosal immunoglobulin A (IgA) response against this lectin can result in fewer recurrent infections.

Cytolytic Mechanisms

- Cytolysis can be undertaken by amebapores, a family of peptides capable of forming pores in lipid bilayers.
- In liver abscess model, trophozoites induced apoptosis via a non-Fas and non-tumor necrosis factor (TNF)- α 1 receptor pathway.
- The amebapores, at sublytic concentrations, also can induce apoptosis.

Role of Cysteine Proteinases

- Cysteine proteinases have been directly implicated in invasion and inflammation of the gut and may amplify interleukin (IL)-1–mediated inflammation by mimicking the action of human IL-1–converting enzyme, cleaving IL-1 precursor to its active form.
- The cysteine proteinases can also cleave and inactivate the anaphylatoxins C3a and C5a, as well as IgA and immunoglobulin G (IgG)

Cell-Mediated Immunity

- Antigen-specific blastogenic responses occur, leading to production of lymphokines, including interferon-delta, which activates the killing of *E histolytica* trophozoites by the macrophages. This killing depends on contact, oxidative pathways, nonoxidative pathways, and nitric oxide (NO).
- Lymphokines, such as TNF- α , are capable of activating the amebicidal activity of neutrophils.
- Incubation of CD8⁺ lymphocytes with *E histolytica* antigens in vitro elicits cytotoxic T-cell activity against the trophozoites.
- During acute invasive amebiasis, T-cell response to *E histolytica* antigens is depressed by a parasite-induced serum factor.

Diagnosis

- Identifying amebic trophozoites or cysts in stool or tissues.
- Pathogenic *E. histolytica* cannot be distinguished morphologically from nonpathogenic *E. dispar* or other species of uncertain pathogenicity.
- Immunoassays detecting *E. histolytica* antigens in stool are sensitive and specific.
- Specific DNA detection assays using polymerase chain reaction (PCR) are available at reference laboratories and offer high sensitivity and specificity.

Diagnosis:

❖ For intestinal infections

- ❖ Enzyme immunoassays of stool.
- ❖ Molecular tests for parasite DNA.
- ❖ Microscopic examination.
- ❖ Serologic testing.

❖ Extraintestinal infections:

- ❖ Diagnosed through imaging, serologic testing, or a therapeutic trial with an amebicide.

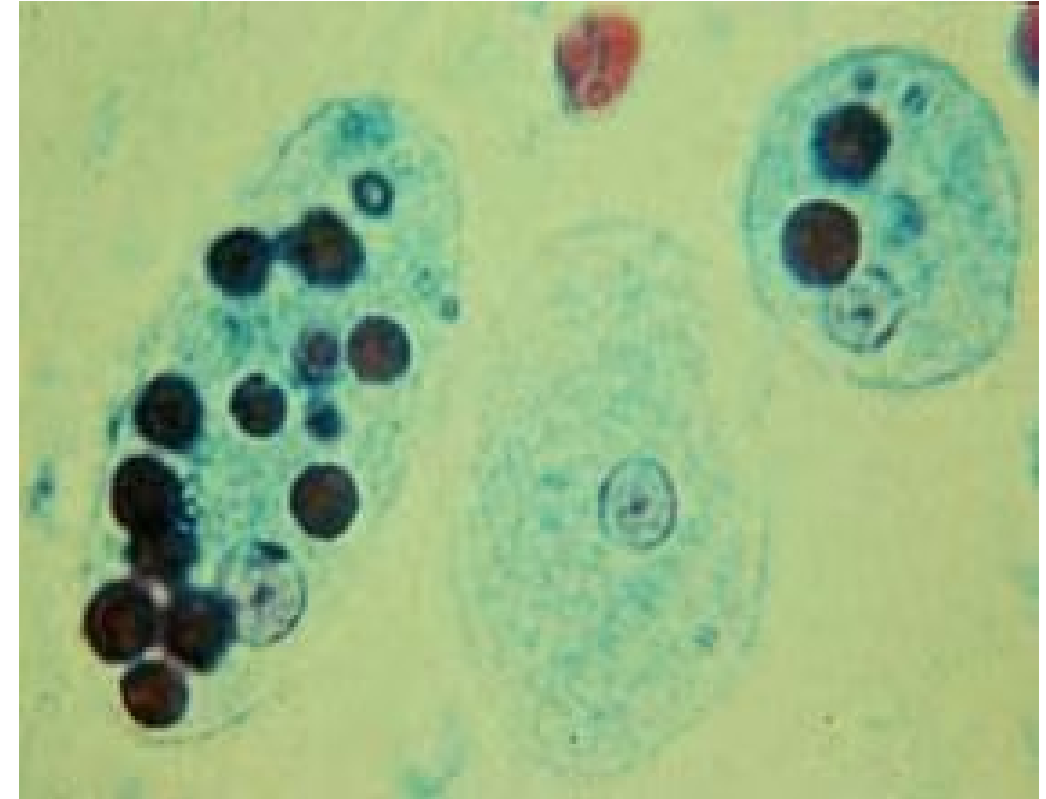
❖ **Non-dysenteric amebiasis:** can be misdiagnosed as irritable bowel syndrome, inflammatory bowel disease, or diverticulitis, while a right-sided colonic mass may be mistaken for cancer or other conditions.

❖ **Amebic dysentery:** can be confused with shigellosis, salmonellosis, schistosomiasis, or ulcerative colitis, but typically presents with less frequent, less watery stools containing tenacious mucus and blood, and fewer white blood cells due to trophozoite lysis.

- Patients with amebic liver abscess often experience right upper quadrant pain and fever, and the condition is more common in younger men exposed to endemic areas.
- Diagnosis require imaging, laboratory tests, and tissue biopsy, including complete blood count (CBC), liver tests, and abdominal CT.
- Amebic liver abscess does not typically cause a left shift in white blood cell counts or elevated serum bilirubin levels, unlike pyogenic liver abscess.

Microscopy

- Involving staining and microscopic examination of fresh stool smears for trophozoites that contain ingested red blood cells
- The presence of intracytoplasmic RBCs in trophozoites is diagnostic of *E histolytica* infection, although some studies have demonstrated the same phenomenon with *E dispar*.
- Examination of 3 stool samples over no more than 10 days can improve the detection.
- Stool leukocytes be found.
- Stool examination findings in patients with amebic liver abscess are usually negative.



Trichrome stain of *Entamoeba histolytica* trophozoites in amebiasis. Two diagnostic characteristics are observed. Two trophozoites have ingested erythrocytes, and all 3 have nuclei with small, centrally located karyosomes.

Culture

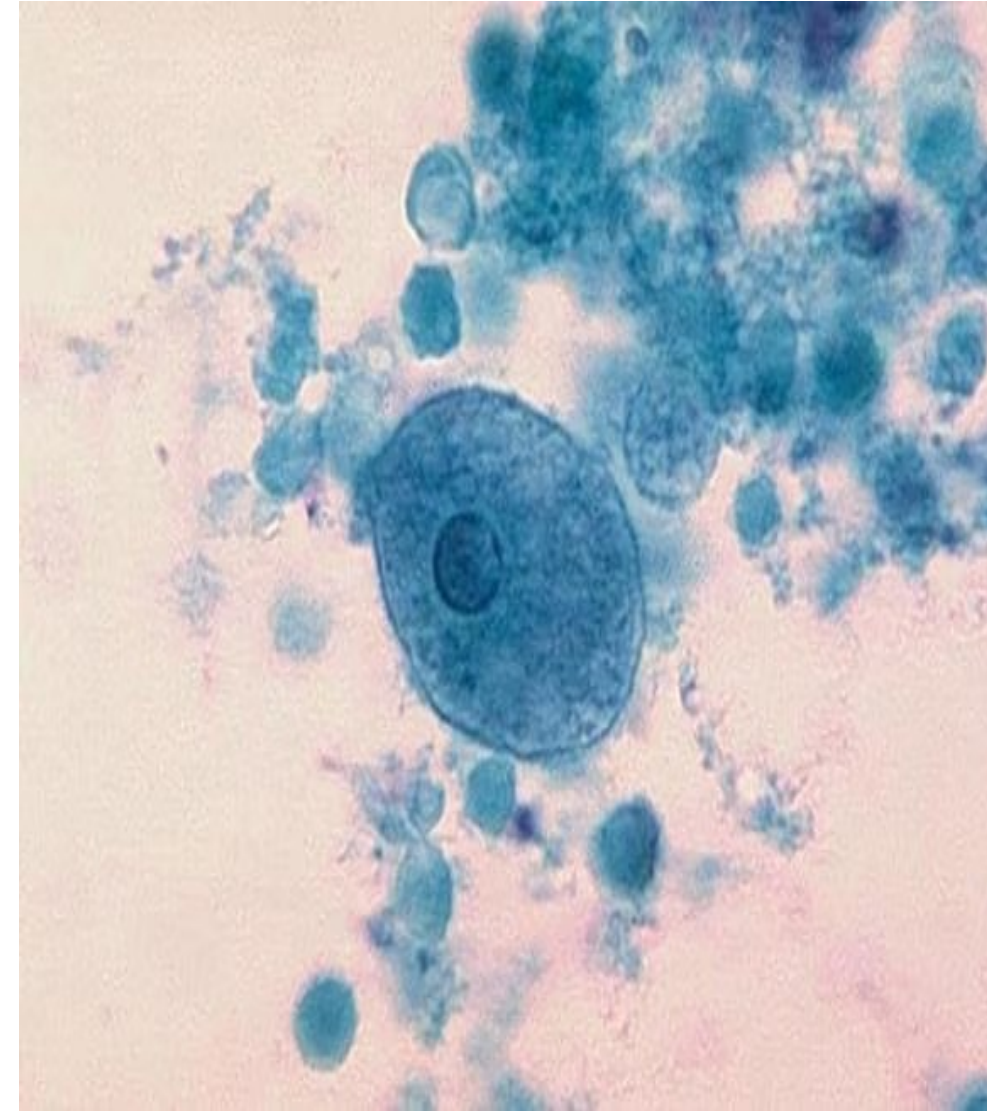
- ❖ Can be performed either with fecal or rectal biopsy specimens or with liver abscess aspirates.
- ❖ Culture has a success rate of 50-70%, but it is technically difficult.
Overall, culture is less sensitive than microscopy.
- ❖ Xenic cultivation, using modified Locke-egg media.

Antigen detection

- (ELISA) is used to detect antigens from *E histolytica* in stool samples. Several kits are commercially available.
- No specific antigen tests are available for the detection of *E dispar* and *E moshkovskii* from clinical samples.
- Real-time PCR is considerably more sensitive than stool antigen detection.

Antibody detection

- Serum antibodies against amebae are present individuals with symptomatic intestinal *E histolytica* infection. Anti-amebic antibodies are present in individuals with liver abscess who have been symptomatic for longer than 1 week.
- Serologic examination should be repeated 1 week later in those with negative test on presentation.
- ELISA, the assay most commonly used worldwide, measures the presence of serum antilectin antibodies (immunoglobulin G [IgG]).
- Immunofluorescent assay (IFA)



Entamoeba histolytica in liver aspirate, trichrome stain. Image courtesy of CDC.

- Indirect hemagglutination assay (IHA) detects antibody specific for *E histolytica*.
- Detection of immunoglobulin M (IgM) antibodies specific for *E histolytica* has been reported

Patient education & prevention

- Avoid drinking contaminated water; use bottled water while traveling if possible
- If local water is to be drunk, purify it by:
 - (a) boiling it for more than 1 minute
 - (b) using 0.22 μm filtration,
 - (c) iodinating it with tetraglycine hydroperiodide
- Avoid eating raw fruits and salads, which are difficult to sterilize; eat only cooked food or self-peeled fruits if possible
- Wash uncooked vegetables and soak them in acetic acid or vinegar for 10-15 minutes



Specimen of the human intestine that was damaged by amoebic ulcer.